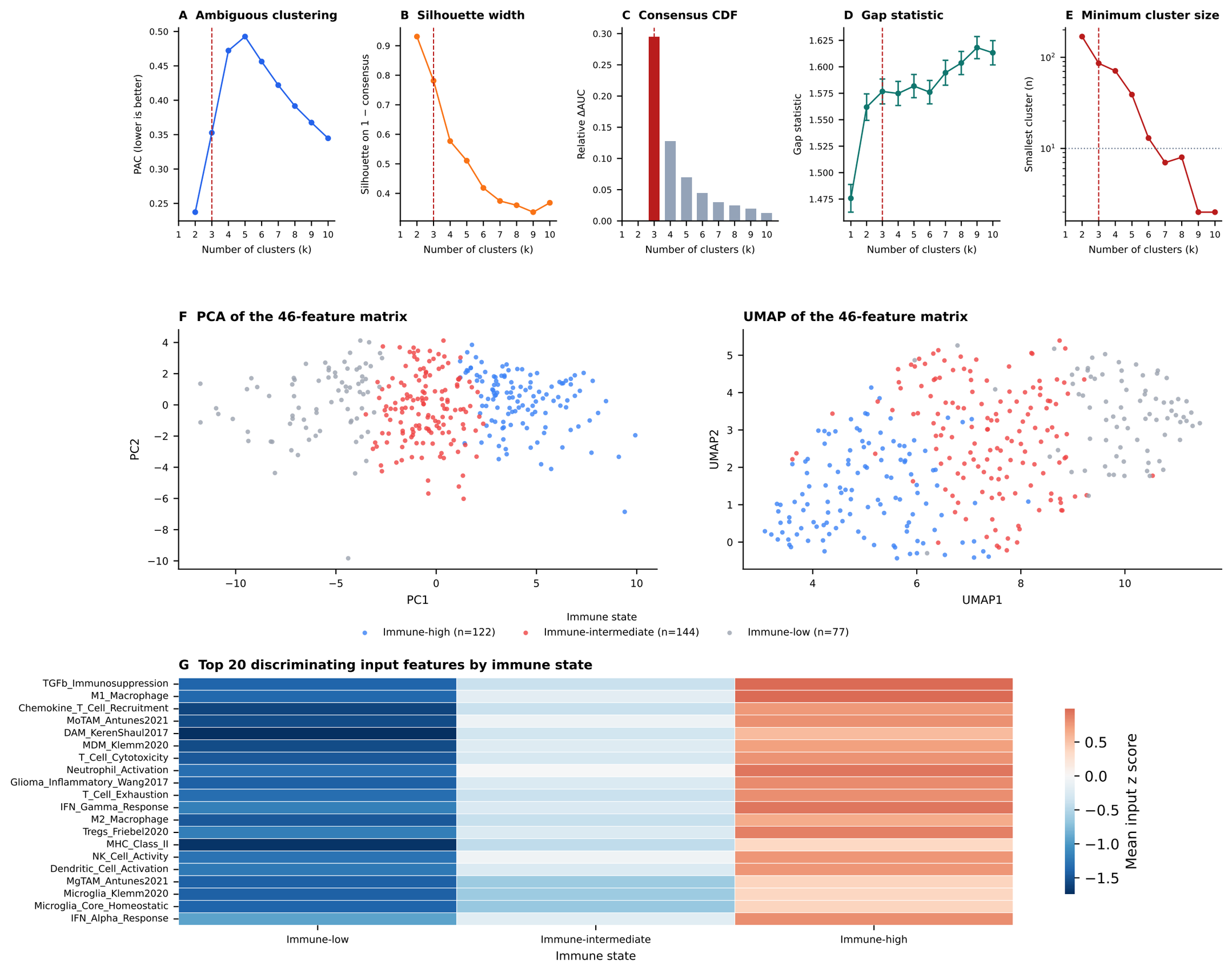

A Ambiguous clustering
PAC (lower is better)
Number of clusters (k)
B Silhouette width
Silhouette on 1 − consensus
Number of clusters (k)
C Consensus CDF
Relative ΔAUC
Number of clusters (k)
D Gap statistic
Gap statistic
Number of clusters (k)
E Minimum cluster size
Smallest cluster (n)
Number of clusters (k)
F PCA of the 46-feature matrix
PC2
PC1
UMAP of the 46-feature matrix
UMAP2
UMAP1
Immune state
Immune-high (n=122)
Immune-intermediate (n=144)
Immune-low (n=77)
G Top 20 discriminating input features by immune state
TGFb_Immunosuppression
M1_Macrophage
Chemokine_T_Cell_Recruitment
MoTAM_Antunes2021
DAM_KerenShaul2017
MDM_Klemm2020
T_Cell_Cytotoxicity
Neutrophil_Activation
Glioma_Inflammatory_Wang2017
T_Cell_Exhaustion
IFN_Gamma_Response
M2_Macrophage
Tregs_Friebel2020
MHC_Class_II
NK_Cell_Activity
Dendritic_Cell_Activation
MgTAM_Antunes2021
Microglia_Klemm2020
Microglia_Core_Homeostatic
IFN_Alpha_Response
Immune-low
Immune-intermediate
Immune-high
Immune state
Mean input z score